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Suffix Tree : Len(n) → n leaves

i.e : banana

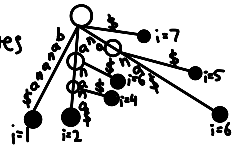
S = banana\$; $|S| = 7$; $n = 7$

S_1 = banana\$

S_2 = anana\$

⋮

S_7 = \$



BWM

How can BWM be used to find location of P in string T?

Sort rotations lexicographically, i.e. T = mississippi\$, n = 12

\$ x
a . . . . y
a . . . . z
b . . . .
⋮
h
⋮

BWT = xyz . . .

0	\$ . . . i	11	\$	
1	i . . . p	10	i\$	
2	i . . . s	7	ippi\$	
3	i . . . s	4	issippi\$	
4	i . . . m	1	:	
5	m . . . \$	0	:	
6	p . . . p	9	:	
7	p . . . i	8	:	
8	s . . . s	6	:	
9	s . . . s	3	:	
10	s . . . i	5	:	
11	s . . . i	2	:	

F = \$iiiimppssss

L = ipssm\$pissii

SA = [11, 10, 7, 4, 1, 0, 9, 8, 6, 3, 5, 2]

C Table F[i]

\$	1	0	0
i	4	1	1-4
m	1	5	5
p	2	6	6-7
s	4	8	8-11

Occ Table

	0	1	2	3	4	5	6	7	8	9	10	11	Tot
\$	0	0	0	0	0	1	1	1	1	1	1	1	1
i	1	1	1	1	1	1	1	2	2	2	3	4	4
m	0	0	0	0	1	1	1	1	1	1	1	1	1
p	0	1	1	1	1	1	1	2	2	2	2	2	2
s	0	0	1	2	2	2	2	2	2	3	4	4	4

LF: new lo : $C[c] + Occ(c, lo-1)$

new hi : $C[c] + Occ(c, hi) - 1$

$c = P[i]$

Backward search → Query $\boxed{✓}$

$L \in NP$ $\boxed{\text{not}}$ polynomial

NP are verifiable in poly
not solvable

Ukkonen → suffix links, skip-count. & global edge : $O(n)$
↳ suffix links & skip-count : $O(n^2)$
↳ No tricks : $O(n^3)$

Halting is not poly.
NP is not bounded by time comp.

$T(n)$ = Exp. # of compare on input size n.

$$T(n) = \begin{cases} T(1) + T(n-1) + O(n) ; \text{ if } 1:n-1 \text{ split} \rightarrow \text{Bad pivot} \\ T(1) + T(n-2) + O(n) ; \text{ if } 1:n-2 \text{ split} \\ \vdots \\ T(1) + T(n-k) + O(n) : \text{ if } 1:n-k \text{ split} \rightarrow \text{Good} \\ T(n-1) + T(1) + O(n) ; \text{ if } n-1:1 \text{ split} \end{cases}$$

n possible choices

$T(n) = T(\text{choose pivot}) + O(n) + T(\text{Right}) + T(\text{Left})$

$= n \sum_{k=1}^{n-1} Pr(p_k) \left[T(k) + T(n-k) \right]$

Analyze Rand. Sel. Alg. when pivot → 90-10% split

$T(n) = T(0.9n) + cn$

$0.9^k \cdot n = 1 \rightarrow k = \log_{0.9}(1/n) = \log_{0.9} n$

$T(n) = cn + c(0.9n) + c(0.9^2 n) + c(0.9^3 n) + \ldots$

$= cn(1 + 0.9 + 0.9^2 \ldots)$

$\sum 0.9_i = 1/(1-0.9) = 10$

$T(n) = 10cn$

$T(0.9n) = T(0.9 + 0.9n) + c(0.9n)$

$T(n) = cn + c(0.9n) + c(0.9^2 n) + \ldots$

$= cn(1 + 0.9 + 0.9^2 + \ldots)$

$\sum_{i=0}^{\infty} 0.9^i = \frac{1}{1-0.9} = 10$

$10cn \rightarrow \boxed{O(n)}$

Subset Sum : $X = \{x_1, x_2, \ldots x_n\}$, target T

Does any sub of X add up to T?

Hard to Solve : 2^n subsets

i.e. $X = \{3, 9, 14, 5, 22, 7\}$, $T = 26$, $S = \{3, 14, 9\}$

Pt 1: $3 \in X$ ✓, $14 \in X$ ✓, $9 \in X$ ✓ : $O(n)$

Pt 2: sum: 0 → 3 → 17 → 26 ✓ $O(n)$

Prove $L' \in P$ if $L \in P$ and $L' \leq_p L$

$L' \leq_p L$ implies exists poly time compute

$x \in L' \rightarrow f(x) \in L$: if $f(x)$ bound by $O(|x|^k)$

a. $y = f(x)$ takes $O(|x|^k)$ time

b. A_L on input y takes $O(|y|^k)$ time

c. return result $A_L(y)$

In rand. sel. expected val. of middle 50% pivot = 2. $E[t] = \frac{1}{p} = \frac{1}{0.5} = 2$

Analyze given expect val. of middle 60% = 10. 60% : $E[t] = \frac{1}{0.6} = \frac{5}{3} \approx 1.667$

Give recurrence relation & upper bound time.

E = 10 → 10cn

larger side ≤ 0.8n → $T(n) = T(0.8n) + 10cn$

$T(n) = 10cn + 10c(0.8n) + 10c(0.8^2 n) \ldots$

$= 10cn(1 + 0.8 + 0.8^2 + \ldots)$

$= \sum_{i=0}^{\infty} 0.8^i = \frac{1}{1-0.8} = 5$

→ $T(n) = 10cn \cdot 5 = 50cn$

$\boxed{O(n)}$

Hamiltonian Path : Graph $G = (V, E)$

Hard to solve : 2! orders

$\pi = [v_3, v_1, v_7, v_2, \ldots, v_n]$

i.e. $\pi[A, C, E, B, D]$;

Pt 1: $|\pi| = n$? → 5 ✓ $O(1)$

Pt 2: Mark each vertex as seen scanning π $O(n)$

Pt 3: $(A \rightarrow C) \in E$ ✓ $(C \rightarrow E) \in E$ ✓ $(E \rightarrow B) \in E$ ✓ $(B \rightarrow D) \in E$ ✓ n-1 checks $\boxed{O(n) ✓}$

Prove Hamiltonion Cycle is NP-Complete

Ham ∈ NP ∧ Ham is NP-Hard

Pt 1. Check $|\pi| = n$
Check π are distinct
Check vertices of G
check consec pair ∈ E
check closing edge ∈ E

Every vertices exactly once and every edge exists btwen every consecutive pair of vertices ✓

Pt 2. We know Ham path → NP Comp ✓

Ham path : $G = (V, E)$, Ham Cycle : $G' = (V', E')$

- Add single new V to G thus $V' = V \cup \{v_{new}\}$
- Add edge connecting v_{new} to every existing V thus $E' = E \cup \{(v_{new}, u) \mid u \in V\}$

Pt 3. Proof of Correct (G Ham path ⟺ G' Ham Cycle

Yes ✓

Time comp. of alg. A that makes polynomial # of calls to a poly. subroutine S.

$\boxed{\text{Polynomial}}$ If alg. A run poly time $O(N^k)$ exclusive calls to S.

Let # of calls to S be bounded by a poly $O(N^p)$

Let subroutine S run in poly time bound $O(M^q)$ → M size of S.

Max M cannot exceed # of ops A. → $M = O(N^k)$

3-SAT : $\phi = (x_1 \vee -x_2 \vee x_3) \wedge (-x_1 \vee x_2 \vee x_4) \wedge (-x_2 \vee -x_4 \vee x_1)$

Does there exist assign. of true/false that makes ϕ true?

i.e. $\alpha = \{x_1 = T, x_2 = F, x_3 = T, x_4 = F\}$

Verify: $C_1 \cdot T \vee T \vee T$ ✓

$C_2 : F \vee F \vee F$ ✗ ← fails

Some radio trans. are too close to run simultaneously.
We want to activate as many as possible.
Is this a poly or NP problem?

Graph = (V, E) → V = vertex (trans), $(u, v) \in E$ intervene trans

Max Ind. Set problem

NP-Complete? : Prove NP membership & NP-hardness

Pt 1: $S \subseteq V$ verify valid ind. set size ≥ k

1. Check $|S| \geq k$ - one pass, $O(n)$
2. For every pair. $(u, v) \in S \times S$, check edge $(u, v) \notin E$ at most $O(n^2)$ poly ✓

P2: 3-SAT : $\phi = (x_1 \vee -x_2 \vee x_3) \wedge (-x_1 \vee x_2 \vee x_4) \wedge \ldots$

3-SAT $\leq_p$ MIS → MIS is NP-Hard $\boxed{\text{NP-Complete}}$

Prove Inde. Set is NP-Complete

Pt 1. IS ∈ NP ; Check $|S| \geq k$ $O(n)$

2. For every pair (u, v) where $u \in S$ & $v \in S$ check $(u, v) \notin E$ $O(n^2)$ NP ✓

Pt 2. NP-Hard

$\phi = C_1 \wedge C_2 \wedge \ldots C_m$, $C_j = (l_{j,1} \vee l_{j,2} \vee l_{j,3})$

$G = (V, E)$ for each C_j and $l_{j,i}$ in it it gives 3m vertices

$V = \{(j, i) : 1 \leq j \leq m, 1 \leq i \leq 3\}$

Two-edge types:

Intra-clause

Connect all 3 literals in same clause

$\forall_j : \{(j,1), (j,2)\}, \{(j,1), (j,3)\}, \{(j,2), (j,3)\} \in$

Conflict

Connect literals across diff clauses

$\{(j, i), (j', i')\} \in E$ if $j \neq j'$ and $l_{j,i} = \neg l_{j', i'}$

Set target k = m

$C_1 = (x_1 \vee -x_2 \vee x_3)$ $C_2 = (-x_1 \vee x_2 \vee x_4)$ $C_3(x_2 \vee -x_3 \vee$

Pt 3: Reduction

ϕ is satisfiable ⟺ G has ind. set size ≥ m

1. $S = \{v_1, v_2, \ldots v_m\}$, S is ind. set size of m
   $|S| = m = k$
2. IS of size m → satisfy
   Assume G has ind. set S → $|S| \geq m$
   1. vertex per clause
   2. consistent assignment
      $\alpha(x_i) = \begin{cases} T & v_j \text{ has } x_i \\ F & v_j \text{ has } -x_i \\ T & \text{free} \end{cases}$
   3. Could α try to set $x_i = T$ and $x_i = F$ simult? S must be ind. ✓
   4. α satisfy ϕ ✓

Vertices - 3m - $O(m)$
Intra - 3m - $O(m)$
Conflict - $9m^2$ - $O(m^2)$

$\boxed{\text{NP-Complete ✓}}$

Random Alg.

$E[X] = \sum x \cdot P(X = x)$ def

$E[aX + b] = a \cdot E[X] + b$ linearity

$E[X + Y] = E[X] + E[Y]$ linearity. always true

$E[XY] = E[X] E[Y]$ only if X, Y ind.

If $X = Geom(p)$: $E[X] = 1/p$ expect trials to first sucess

Rand. Select (A, lo, hi, k):

```
if lo == hi : return A[lo]
pivot ← A[rand ind. in [lo, hi]]
q ← Partition (A, lo, hi, pivot)
rank ← q - lo + 1
if k == rank: return A[q]
if k < rank: return Rand.Select(A, lo, q-1, k)
else : return Rand.Select (A, q+1, hi, k-rank)
```

NP-Complete

Pt 1 : X ∈ NP

- Show certificate
- Verify poly time
- Verify accept certificate

Pt 2 : X is NP-Hard

- Pick NPC problem
- Show $L' \leq X$ (poly time reduction)
- Prove Correctness (Yes ⟺ Yes)

Rec. Analysis Worst case → 90/10 split

$T(n) = T(0.9n) + cn$

Unroll : $T(n) = cn + c(0.9n) + c(0.9^2 n) + \ldots$

$= cn \cdot \sum 0.9^i$

$= cn \cdot \frac{1}{1-0.9}$

$= 10cn = O(n)$

Expected case

$P(50\%) = 0.5 \rightarrow E[\text{trials}] = 2$

$P(60\%) : 0.6 \rightarrow E[\text{trials}] = 5/3$

Good pivot guarentee : worst split ≤ 0.75n (middle 50%)

Recurrence. $T(n) = T(0.75n) + 2cn$

Unroll : $T(n) = 2cn \cdot \sum 0.75^i = 2cn \cdot 4 = 8cn = O(n)$

Any A : $T(n) = T(an) + cn$

$= cn \cdot \sum a^i$

$= cn \cdot \frac{1}{1-a}$

$= O(n)$ ✓

Exp : $O(n)$ Best : $O(n)$

Worst : $O(n^2)$ Space : $O(\log n)$